

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

GCSE MATHEMATICS

F

Foundation Tier Paper 2 Calculator

Thursday 7 June 2018

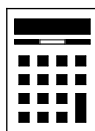
Morning

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a calculator
- mathematical instruments.



Instructions

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.

Advice

- In all calculations, show clearly how you work out your answer.

For Examiner's Use

Pages	Mark
2–3	
4–5	
6–7	
8–9	
10–11	
12–13	
14–15	
16–17	
18–19	
20–21	
22–23	
24–25	
TOTAL	



J U N 1 8 8 3 0 0 2 F 0 1

Answer **all** questions in the spaces provided

Do not write
outside the
box

- 1** Circle the expression that can be written as $2y$ **[1 mark]**

$y + y$

y^2

$2 + y$

$y \times y$

- 2** Circle the decimal that is greater than $\frac{3}{10}$ and less than $\frac{2}{5}$ **[1 mark]**

0.32

0.035

0.4

0.24

- 3** What is 625 as a power of 5 ?
Circle your answer. **[1 mark]**

5^3

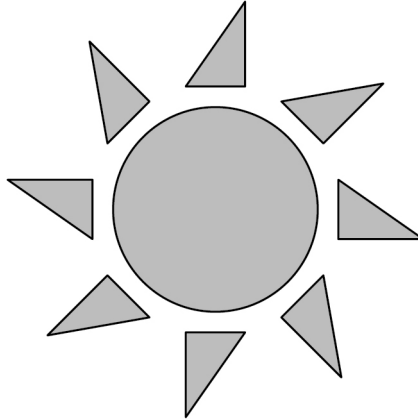
5^4

5^5

5^{125}



- 4 Circle the order of rotational symmetry of this drawing.



[1 mark]

0

2

4

8

- 5 Work out the value of $3^6 - \sqrt{841}$

[2 marks]

Answer _____

Turn over for the next question



- 6** Gemma has four groups of friends on a social media site.
The table shows the number of friends in each group.

Group	Number of friends
Family	8
Netball	8
School	26
Guides	11

- 6 (a)** Which group is the mode?

[1 mark]

Answer _____

- 6 (b)** Gemma wants a pictogram to show the information.
She has drawn the first two rows.
Complete the pictogram.
Remember to complete the key.

[3 marks]

Key: ○ represents _____ friends

Family	○ ○
Netball	○ ○
School	
Guides	



- 7 e is 3 **more** than d .
 f is 5 **less** than d .

- 7 (a) Write an expression for e in terms of d .

[1 mark]

Answer _____

- 7 (b) Write an expression for f in terms of d .

[1 mark]

Answer _____

- 7 (c) Work out $e - f$
Simplify your answer.

[2 marks]

Answer _____

Turn over for the next question



8

The numbers 1 to 12 are put in a grid.

2, 4, 5, 7, 10 and 12 are shown.

		5	10
12			
4			
7		2	

Each of the four sides of the grid must add up to 26

Complete the grid using the numbers

1, 3, 6, 8, 9 and 11

[3 marks]

9

In this question, use

1 foot = 12 inches

1 inch = 2.5 centimetres

Change 5 feet 8 inches to centimetres.

[3 marks]

Answer _____ cm

10

Which of these numbers has **exactly four** factors?

Circle your answer.

[1 mark]

4

8

12

16

Turn over for the next question**Turn over ►**

11

Nick has a 6-digit code.

He remembers it as three 2-digit numbers.

The first number is between 10 and 20

The second number is 3 times the first number.

The third number is 5 times the first number.

All six digits are **different**.

Work out the code.

[3 marks]

Answer _____

12

How many minutes are there in $5\frac{1}{4}$ hours?

Circle your answer.

[1 mark]

315

325

515

525



- 13** Here is a formula for the amount of water needed to cook rice.

$$w = 1.5r + 0.5$$

w is the number of cups of water needed

r is the number of cups of rice to be cooked

- 13 (a)** How many cups of water are needed to cook 7 cups of rice?

[2 marks]

Answer _____

- 13 (b)** How many cups of rice can be cooked with 20 cups of water?

[3 marks]

Answer _____

Turn over for the next question



- 14 (a)** Use your calculator to work out $9.95^2 \times 29.8$

Give your answer as a decimal.

Write down your full calculator display.

[1 mark]

Answer _____

- 14 (b)** Is your answer to part (a) sensible?

Use approximations to decide.

You **must** show your working.

[3 marks]

Tick a box.

☐

Sensible

☐

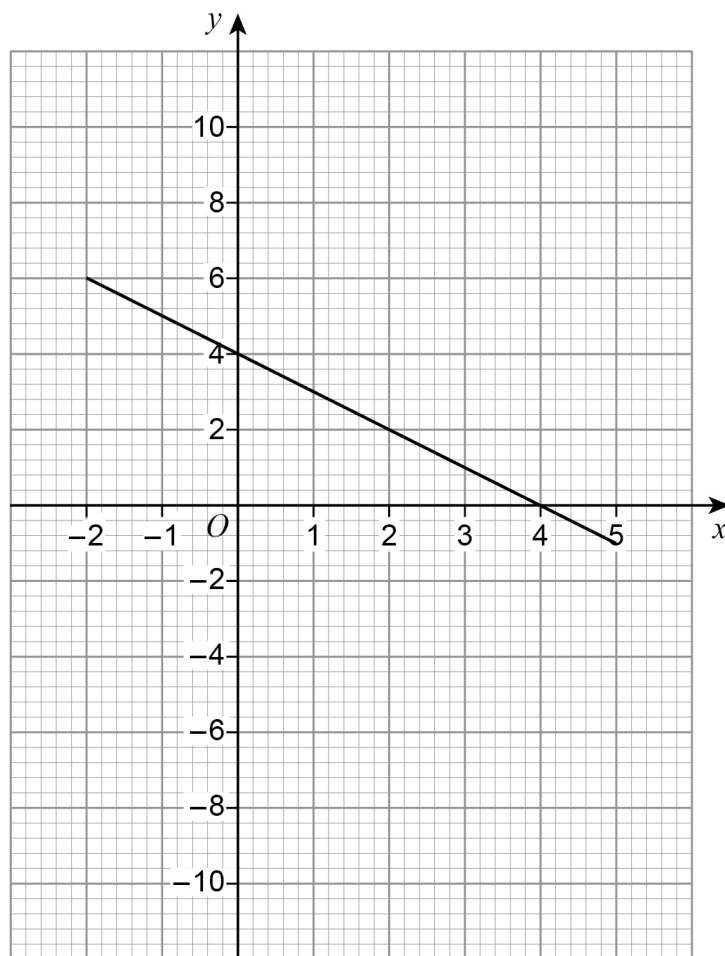
Not sensible



15 The graph of $y = 4 - x$ for values of x from -2 to 5 is shown on the grid.

15 (a) On the grid, draw the graph of $y = 2x - 5$ for values of x from -2 to 5

[3 marks]



15 (b) Use your graph to solve $2x - 5 = 4 - x$

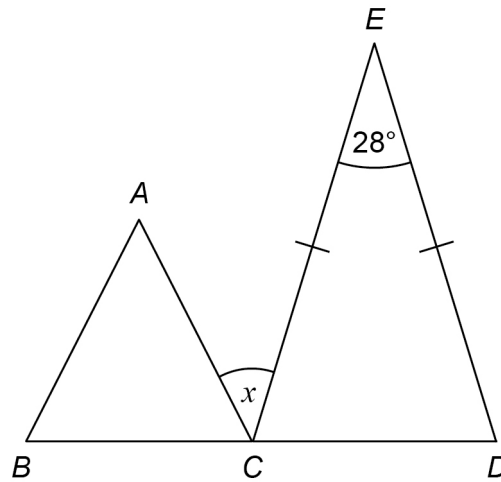
[1 mark]

$x =$ _____



- 16 (a)** BCD is a straight line.
Triangle ABC is equilateral.
 $CE = DE$

Not drawn
accurately



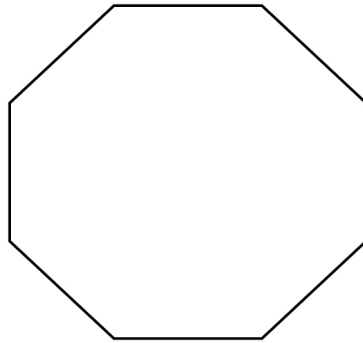
Work out the size of angle x .

[4 marks]

Answer _____ degrees



- 16 (b)** Amba is working out the size of an **interior** angle of a regular octagon.



Not drawn
accurately

Her method is Interior angle = $360 \div 8$

Is her method correct?

Tick a box.

☐

Yes

☐

No

Give a reason for your answer.

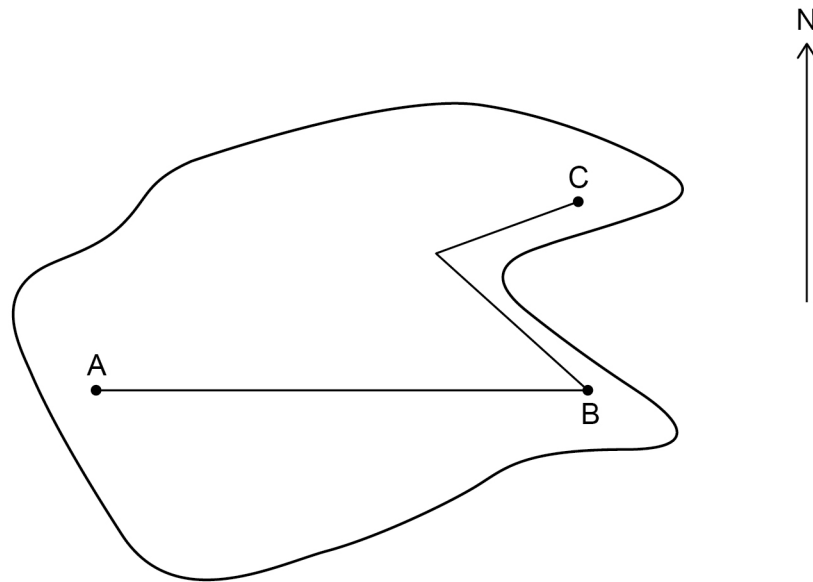
[1 mark]

Turn over for the next question



- 17** Here is a map of an island with cities A, B and C.
The straight lines represent roads.

Scale: 1 cm represents 200 km



- 17 (a)** A is due West of B.
Write down the bearing of A from B.

[1 mark]

Answer _____ °



17 (b) Umar drives from A to B on the route shown.

Kaz drives from B to C on the route shown.

Use the map to work out how much further Umar drives than Kaz.

You **must** show your working.

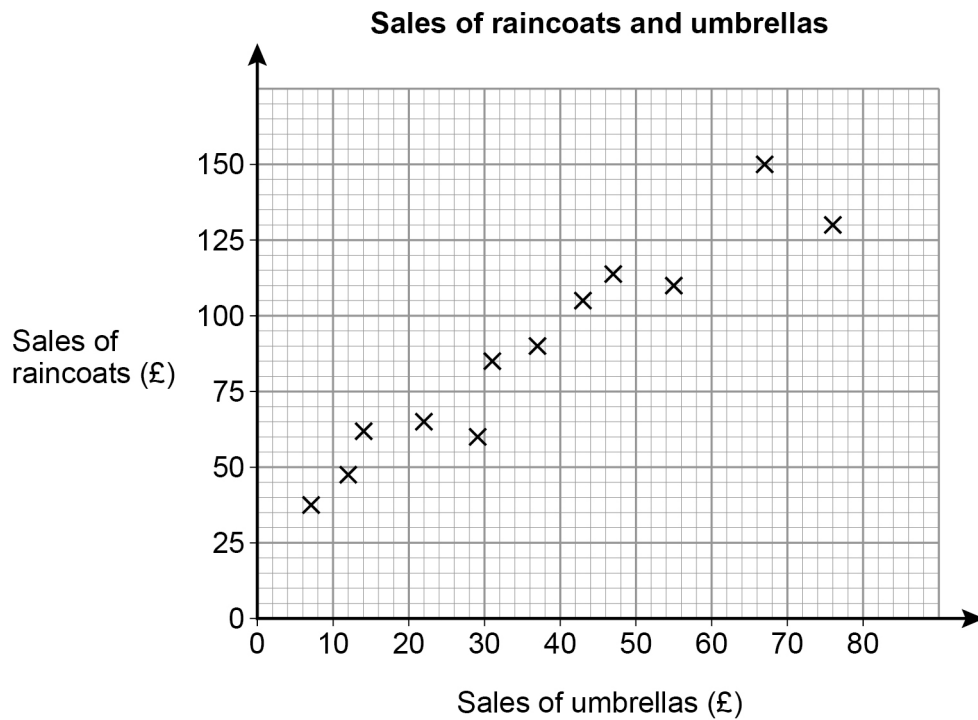
[5 marks]

Answer _____ km

Turn over for the next question



- 18** A shop sells raincoats and umbrellas.
The scatter graph shows the monthly sales for 12 months.



- 18 (a)** Write down the type of correlation shown by the graph.

[1 mark]

Answer _____

- 18 (b)** The manager expects the sales of umbrellas next month to be £60
Draw a line of best fit to estimate the sales of raincoats next month.

[3 marks]

Answer £ _____



- 19** Multiply out $x(x - 4)$
Circle your answer.

[1 mark]

$x^2 - 4$

$2x - 4$

$x^2 - 4x$

$-3x^2$

- 20** $a : b = 5 : 2$

How many times larger is a than b ?
Circle your answer.

[1 mark]

0.4

1.5

2.5

3



21 (a) A circle has radius 4.2 cm

Work out the length of the circumference.

Give your answer to 1 decimal place.

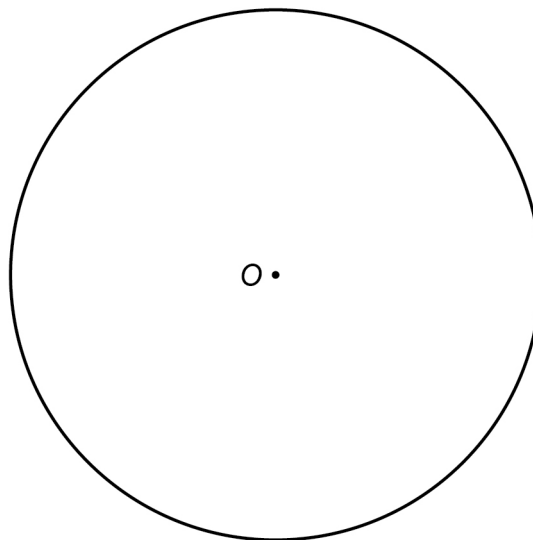
[3 marks]

Answer _____ cm

21 (b) The circle below has centre O .

Draw a sector on the circle.

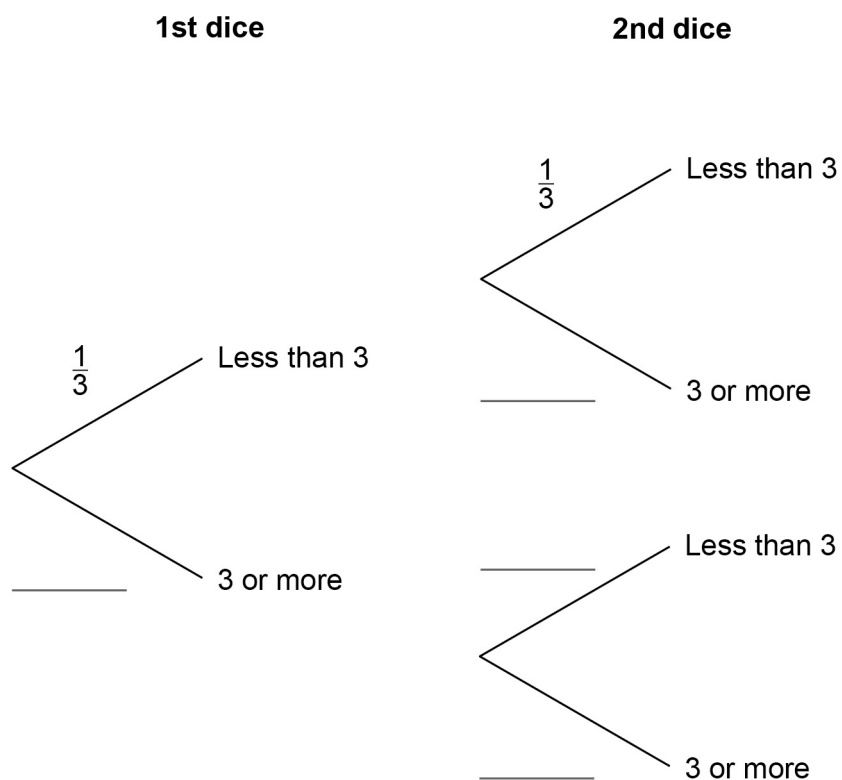
[1 mark]



22 Two ordinary fair dice are rolled.

22 (a) Complete the tree diagram.

[1 mark]



22 (b) Work out the probability that **both** dice land on a number less than 3

[1 mark]

Answer _____

Turn over for the next question



23

Match each sequence to its description.
One has been done for you.

[4 marks]

1 1 2 3 5 8

Arithmetic progression

1 2 4 8 16 32

Geometric progression

1 2 3 4 5 6

Fibonacci sequence

1 3 6 10 15 21

Triangular numbers

1 4 9 16 25 36

Cube numbers

1 8 27 64 125 216

Square numbers



24

The table shows information about the population of a city.

Population in 2001	Population in 2011
420 000	480 000

Liam claims,

“From 2011 to 2021 the population of the city will increase by the same percentage as from 2001 to 2011”

He works out,

$$\begin{aligned}\text{population increase from 2001 to 2011} &= 480\,000 - 420\,000 \\ &= 60\,000\end{aligned}$$

$$\begin{aligned}\text{population in 2021} &= 480\,000 + 60\,000 \\ &= 540\,000\end{aligned}$$

Does the population of 540 000 match his claim?

You **must** show your working.

[3 marks]

Answer _____



- 25** On three days, Ali throws darts at a target.
Here are his results.

	Number of throws	Number of hits	Number of misses
Monday	20	15	5
Tuesday	30	22	8
Wednesday	40	17	23
Total	90	54	36

- 25 (a)** Work out **two** different estimates for the probability of Ali hitting the target.

[2 marks]

Answer _____ and _____

- 25 (b)** Which of your two answers is the better estimate for the probability of Ali hitting the target?

Give a reason for your answer.

[1 mark]

Answer _____

Reason _____



26

Theo starts with savings of £18

James starts with no savings.

Each week from now,

Theo will save £4.50 and James will save £4

In how many weeks will Theo and James have savings in the ratio 15 : 8 ?

[3 marks]

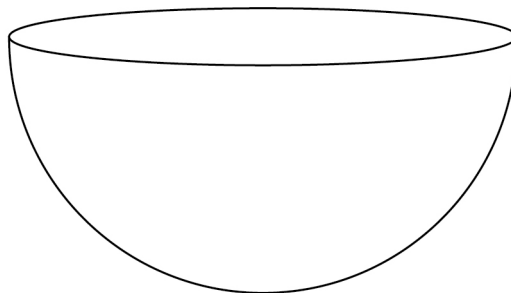
Answer _____

Turn over for the next question

27

Volume of a sphere = $\frac{4}{3}\pi r^3$ where r is the radius

A container is a hemisphere of radius 30 cm



Sand fills the container at a rate of 4000 cm^3 per minute.

Does it take **less than** a quarter of an hour to fill the container?

You **must** show your working.

[3 marks]

Answer _____



28 The length of each side of a regular pentagon is 8.4 cm to 1 decimal place.

28 (a) Complete the error interval for the length of one side.

[2 marks]

_____ cm $\leq$ length < _____ cm

28 (b) Complete the error interval for the perimeter.

[1 mark]

_____ cm $\leq$ perimeter < _____ cm

END OF QUESTIONS

